

Neural Strike: RCA Cybersecurity Final Project

Developed for Rwanda Coding Academy - March 2026

This project demonstrates a multi-stage "Trojan Horse" attack simulation, featuring a custom Dino Jump game, a persistent interactive reverse shell, and a dedicated security cleanup tool.

1. Installation & Setup

For the Attacker (Teacher/Attacker Machine)

Requirements: Python 3.x installed.

Steps:

- Copy attacker.py to your machine.
- Run: python attacker.py
- Console displays listening message.

Network: Same Wi-Fi as target machine.

For the Target (Victim Machine)

Requirements: Windows 10/11.

Steps:

- Run NeuralStrike.exe.

Trigger: Automatically connects to attacker IP or use F1 override.

2. The Gaming Process (User Experience)

- Disclaimer via tkinter modal.
- Dependency staging with progress bar.
- Dino Jump game runs while shell connects in background.

4. Attack Methodology & Persistence

- Interactive reverse shell.
- Registry Run Key persistence.
- Survives system restart.

5. Detection & Prevention

- Monitor registry changes.

- Block unusual network traffic.
- Use antivirus behavioral analysis.

6. Ethical Considerations

- Educational use only.
- Consent required.
- Cleanup tool provided.